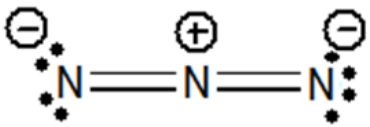
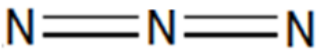
Lewis	#EG	EGG	shape	polar?
	2			No
		linear	linear	

Lewis draw the Lewis structure

#EG Count the # of Electron groups around central atom

Remember: multiple bonds, single bonds and each lone pair around the central atom all count as one EG

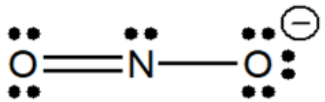
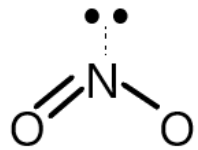
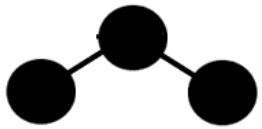
EGG draw the electron group geometry

shape draw the molecular shape

polar? Is the species polar

(if multiple Lewis structures exist: Is the resonance hybrid polar?

How about Nitrite?

Lewis	#EG	EGG	shape	polar?
	3			Yes

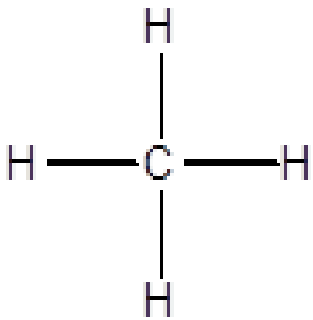
Trigonal planar **bent**

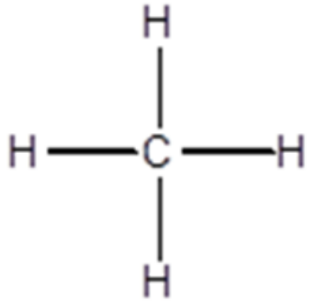
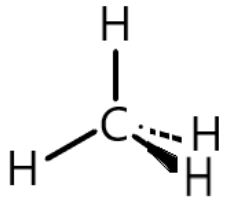
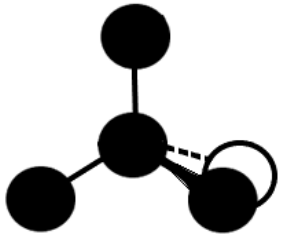
Note that we place the three electron groups as far away as possible to minimize repulsion (trigonal planar EGG) but the molecular shape is derived only from bonding electron pairs. We don't "see" the lone pairs, but they do take up equal space!

So in trigonal planar EGG we can get two molecular shapes:

Trigonal planar
Bent (120 ^{0})

How about Methane?

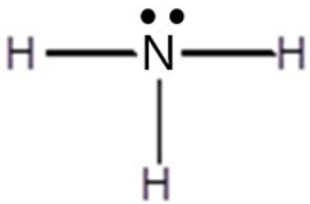
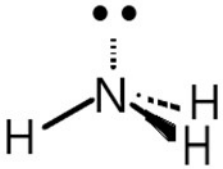


Lewis	#EG	EGG	shape	polar?
	4			No

tetrahedral tetrahedral

The fourth electron group presses out in the third dimension to minimize Repulsion. **Note that the fourth H atom (the one pointing into the plane of the paper) is given a white color for clarity only**.

How about Ammonia?

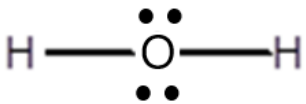
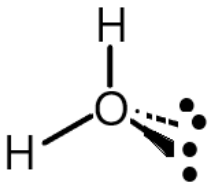
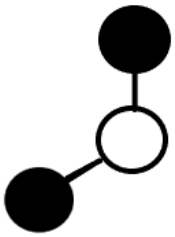
Lewis	#EG	EGG	shape	polar?
	4			Yes

tetrahedral

trigonal pyramidal (tripod)

Note that in the "ball and stick" perspective of the molecular shape I drew the hydrogen atom pointing into the plane of the paper in white so that the drawing would appear mor three-dimensional

Why is water a liquid?

Lewis	#EG	EGG	shape	polar?
	4			Yes
		Tetrahedral	Bent	

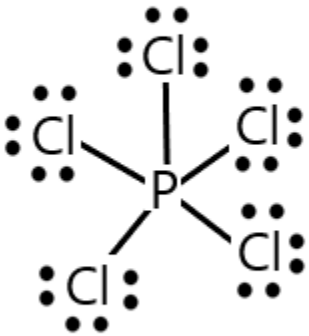
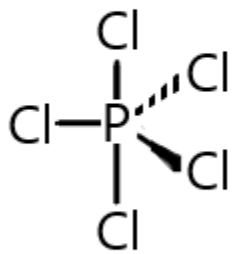
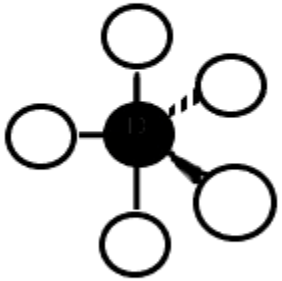
So in tetrahedral EGG we can get three molecular shapes:

Tetrahedral

Trigonal pyramidal (tripod)

Bent (109 ^{0})

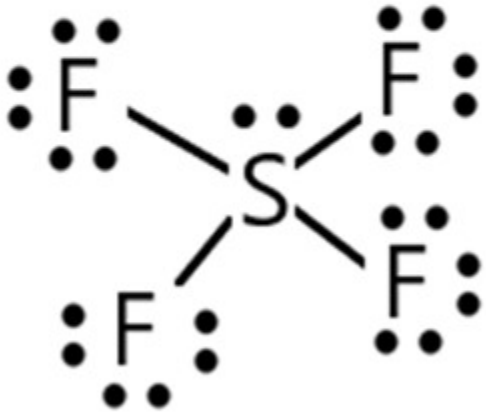
Phosphorus pentachloride

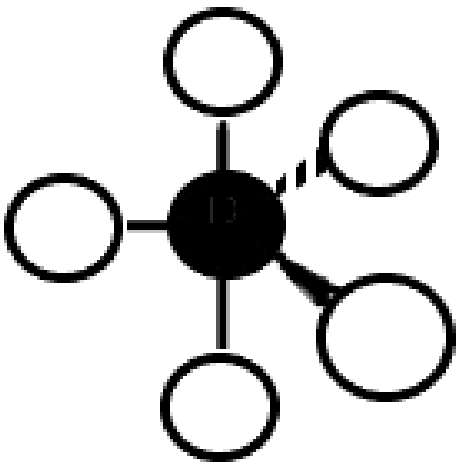
Lewis	#EG	EGG	shape	polar?
	5			No

Trigonal Bipyramidal

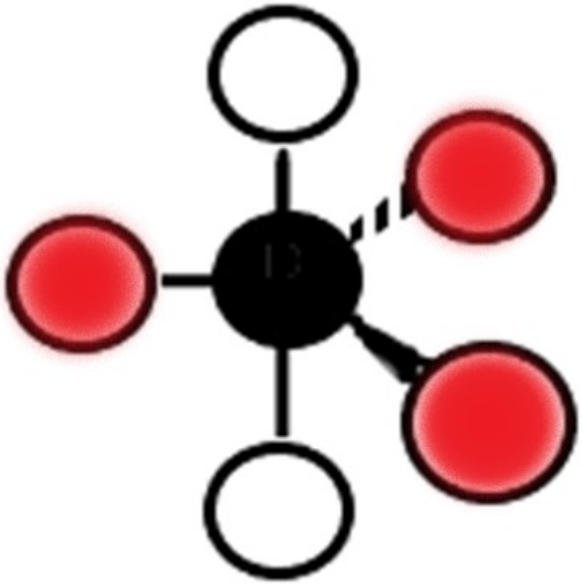
trigonal bipyramidal

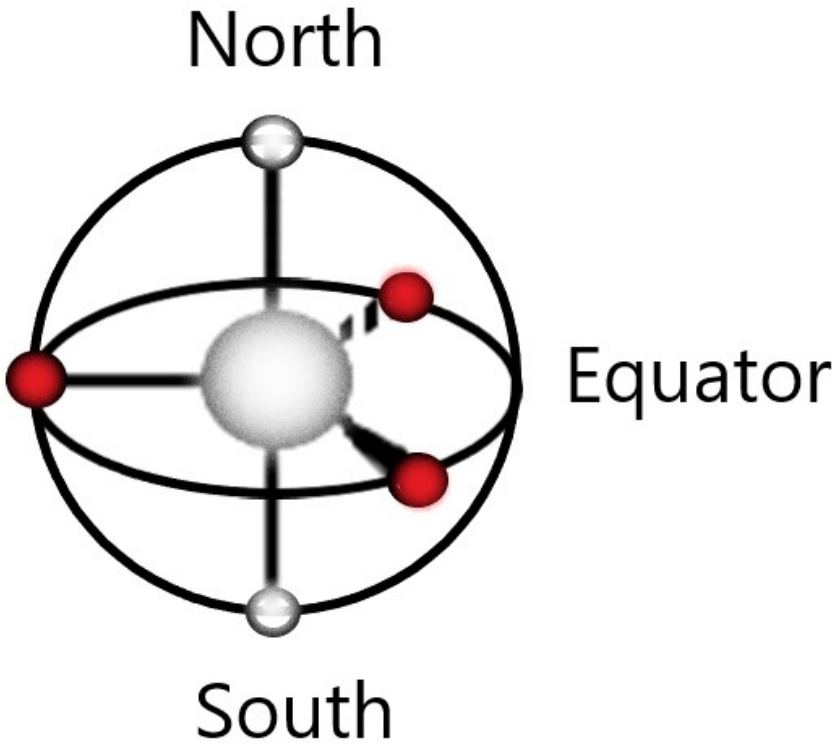
Sulfur tetrafluoride

Lewis	#EG
	5

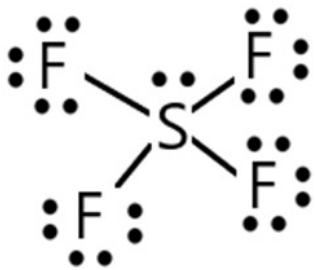
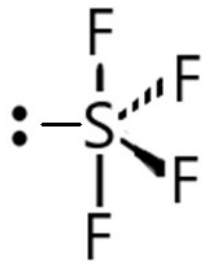


Where do we place the lone pair???
In a trigonal bipyramid there are two different sites!





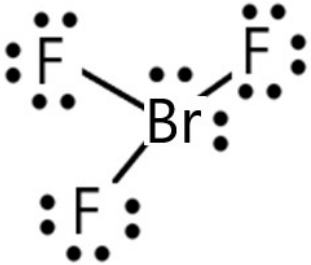
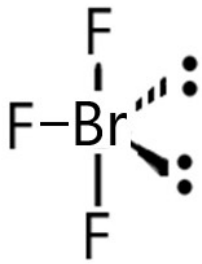
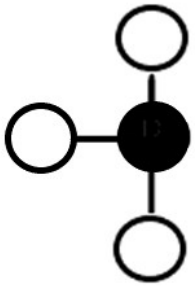
There two AXIAL sites and three EQUATORIAL sites
Lone pairs need more space. In TBP EGG always place lone pairs in the equatorial positions!!!!!

Lewis	#EG	EGG	shape	polar?
	5			Yes

Trigonal Bipyramidal

See-saw (Canada Goose)

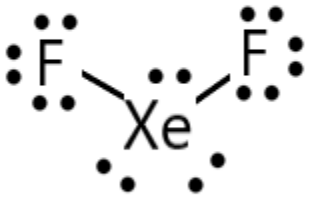
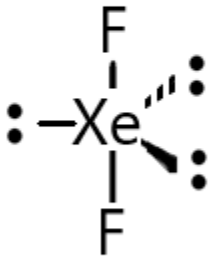
Bromine trifluoride

Lewis	#EG	EGG	shape	polar?
	5			Yes

Trigonal Bipyramidal

T-shaped

Xenon difluoride

Lewis	#EG	EGG	shape	polar?
	5			No

Trigonal Bipyramidal

Linear

So in trigonal pyramidal EGG we can get four molecular shapes:

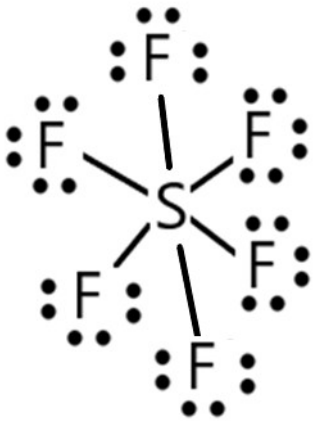
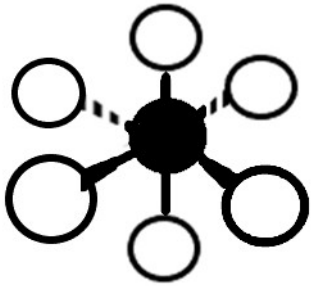
Trigonal bipyramidal

see-saw (Canada goose)

T-shaped

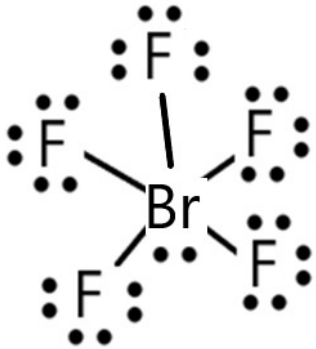
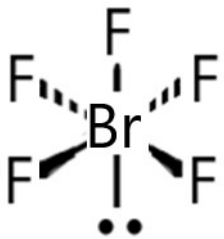
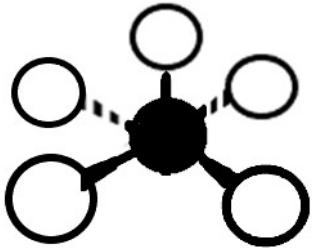
Linear

Sulfur hexafluoride

Lewis	#EG	EGG	shape	polar?
	6			No

Octahedral Octahedral

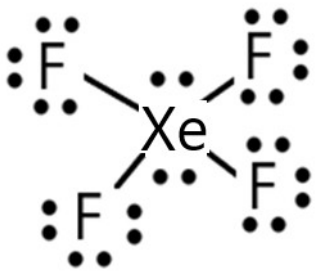
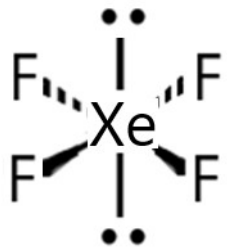
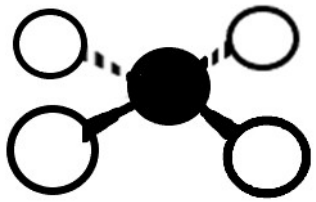
Bromine pentafluoride

Lewis	#EG	EGG	shape	polar?
	6			Yes

Octahedral

square pyramidal

Xenon tetrafluoride

Lewis	#EG	EGG	shape	polar?
	6			No

Octahedral

square planar

So in octahedral EGG we can get three molecular shapes:

Octahedral

Square pyramidal

Square planar